Analysis of ACAS-Xu Systems for Future Compressions and Optimizations using Artificial Neural Networks

ICCAS 2024 Conference

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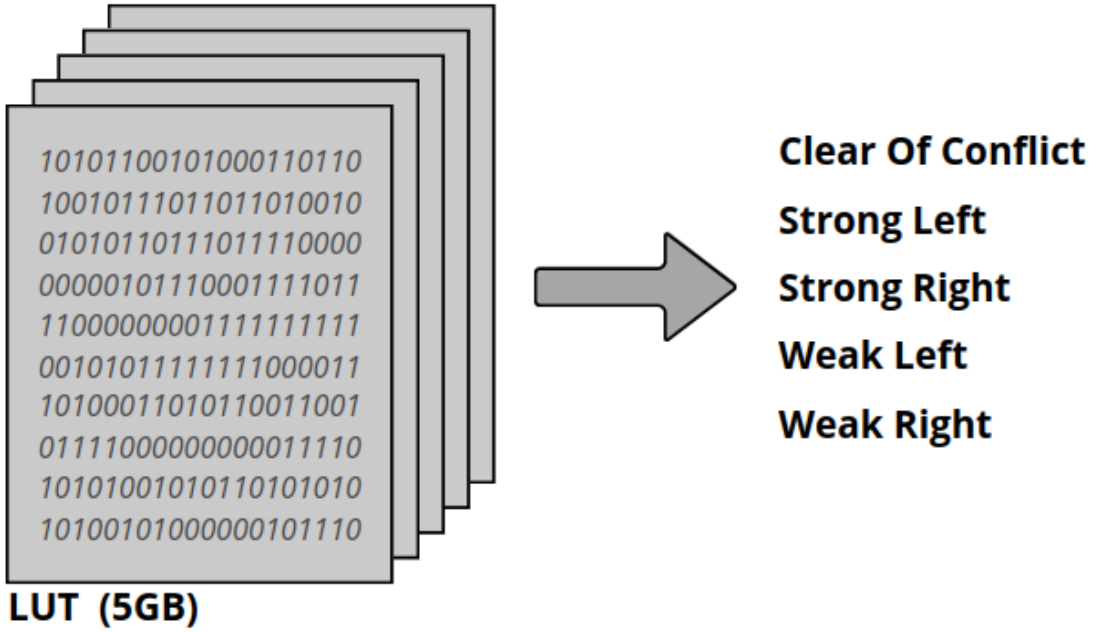
12 mai 2024

Context & Introduction

Anti-collision systems for air traffic are complex but essential.

- TCAS since the 80s.
- ACAS-X new generation :
  $\rightarrow$ ACAS-**Xa**
  $\rightarrow$ ACAS-**Xo**
  $\rightarrow$ ACAS-**Xu**

Context & Introduction



These systems are using Look Up Tables (LUT) for action selection.
The objective is to update today technologie : LUT $\rightarrow$ Neural Network

Then the challenge is to find the precise neural network which learn the LUT data in order to approximate the fonctionement of the ACAS-Xu.

Plan

I - ACAS-Xu Presentation and Zoom on important fonctions.

II - Approximations & Hypothesis for our work.

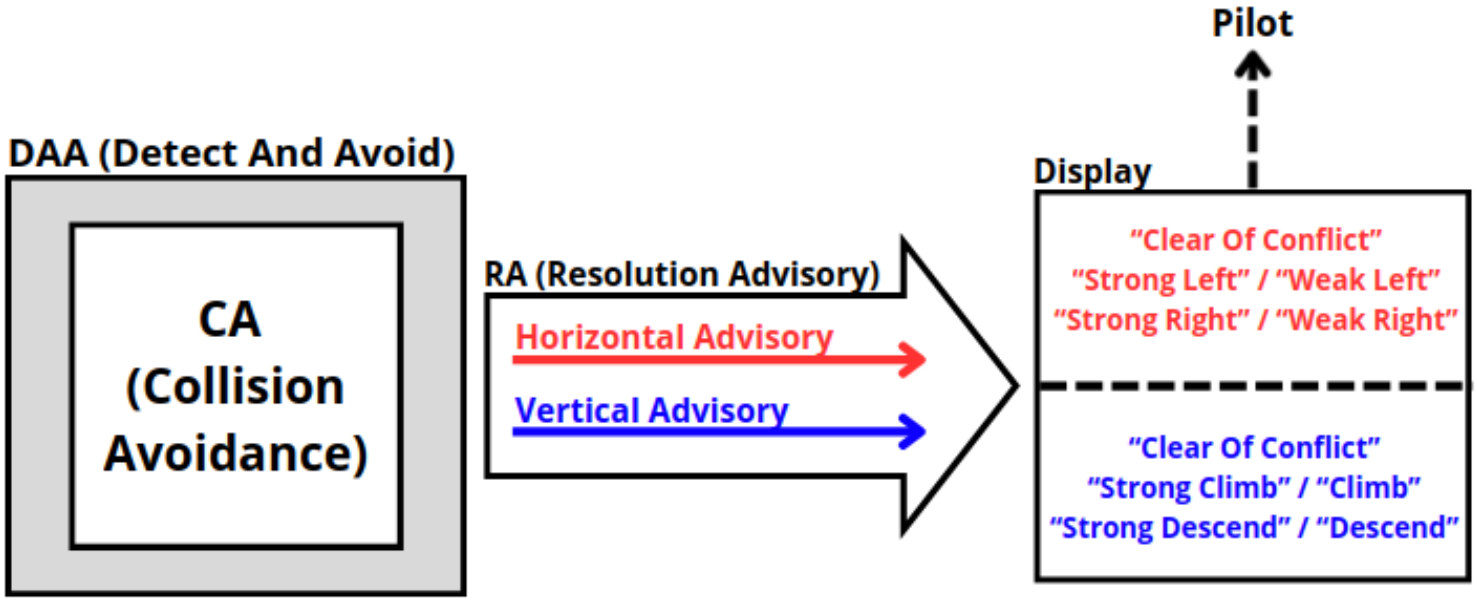
III - Our Neural Network.

I - ACAS-Xu Presentation and Zoom on important fonctions.

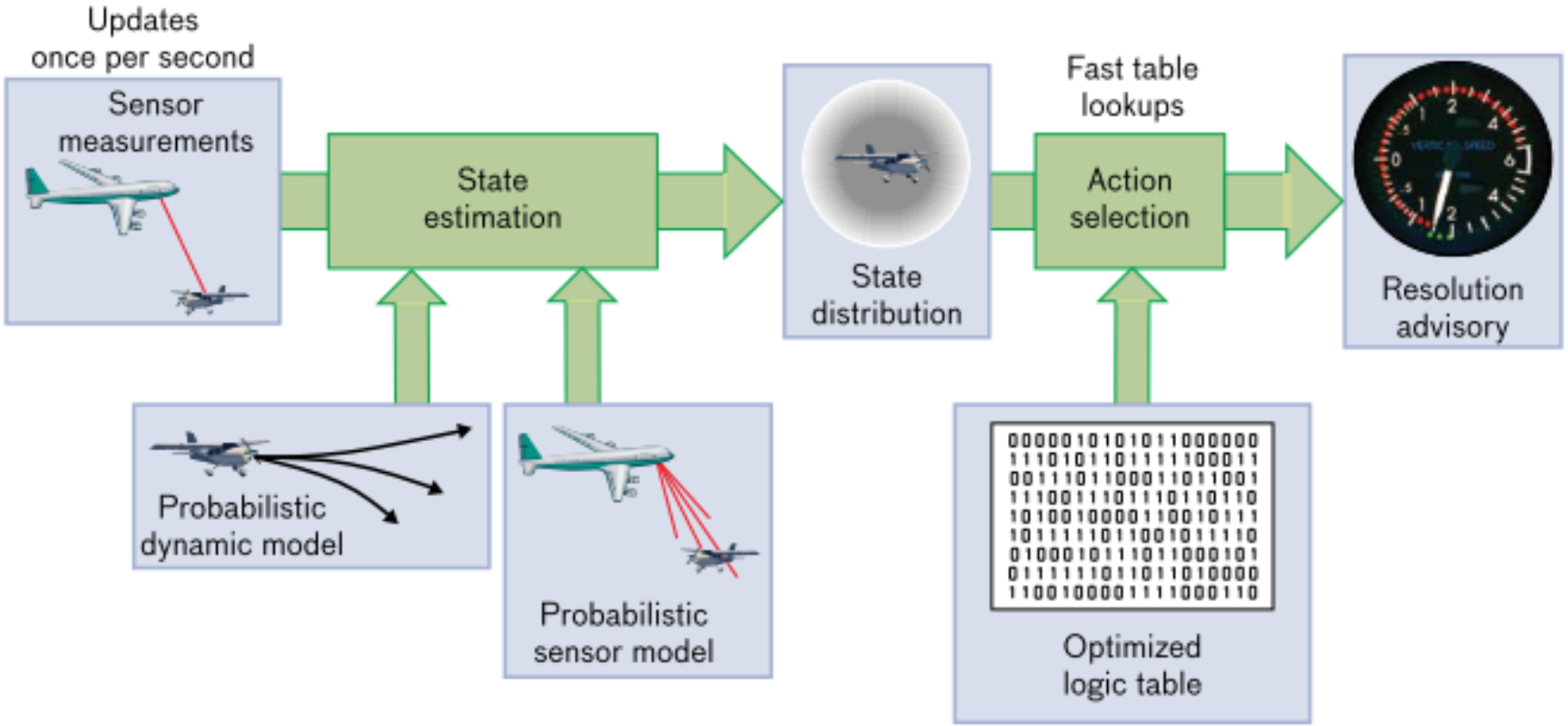
ACAS Presentation

This is a Detect And Avoid (DAA) System.
Every DAA got two main functions : Remain Well Clear (RWC) and Collision Avoidance (CA).

In the of ACAS-XU, 2 independent Resolution Advisory (RA) : Horizontal & Vertical.



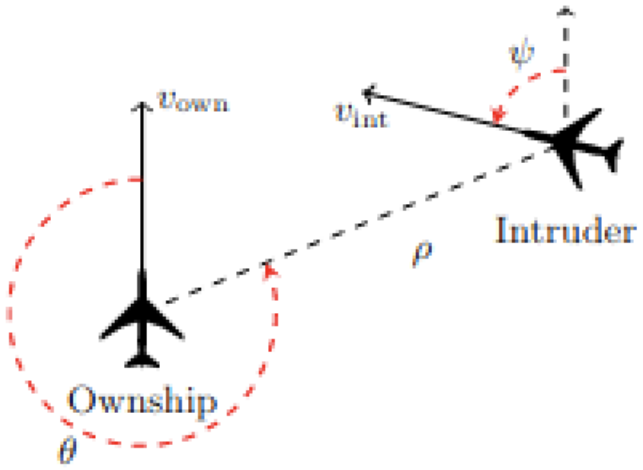
ACAS-X Global Overview



The Look Up Table

Big Array of 11 columns.

- 6 for the flight and geometric variables.
- 5 for the costs of each possible action (COC, SL, SR, WL, WR)



τ	θ	ψ	V_own	V_int	ρ	Cost_COC	Cost_SL	Cost_SR	Cost_WL	Cost_WR
0.0	-3.14	-3.14	50.0	100	499	24017	24017	24081	24001	24065
32.2	-3.14	0.0	50.0	142	3024	22913	22912	23040	22881	23008
41.7	0.0	3.14	87.6	807	45202	22645	22644	22772	22613	22741
101	3.14	1.5	87.6	1069	763118	20844	20832	21340	20700	21208

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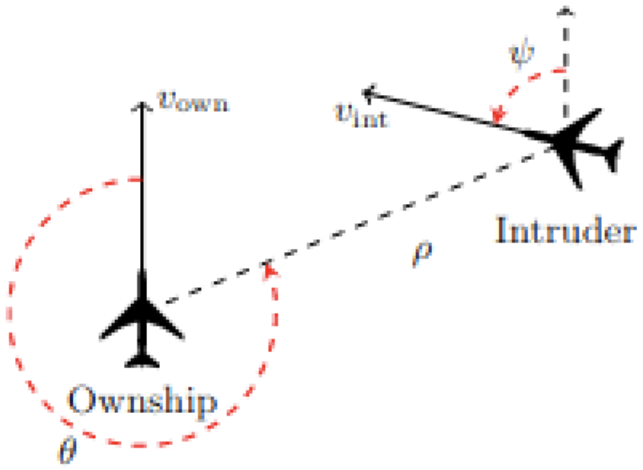

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The Action Selection

- **Vertical Action**

  Multi-linear Interpolation on the Vertical LUT from the state of the problem.
  The Final Action is the one with the lowest cost.

τ	θ	ψ	V_own	V_int	ρ	Cost_COC	Cost_SCL	Cost_SDES	Cost_CL	Cost_DES
0.0	-3.14	-3.14	50.0	100	499	24017	24017	24081	24001	24065
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The Action Selection

- **Horizontal Action**

  Nearest Neighbor approach on the Horizontal LUT from the state of the problem.
  The action with the lowest cost is the "instantaneous action".
  The **Final Action** is found with a lookahead

τ	θ	ψ	V_own	V_int	ρ	Cost_COC	Cost_SL	Cost_SR	Cost_WL	Cost_WR
0.0	-3.14	-3.14	50.0	100	499	24017	24017	24081	24001	24065
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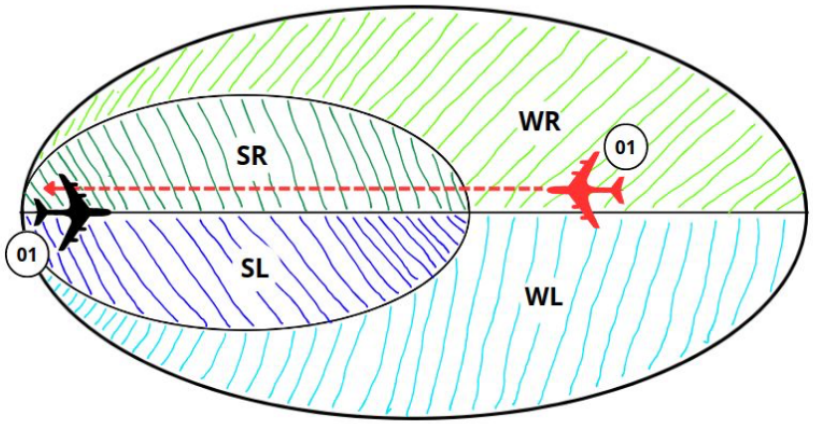
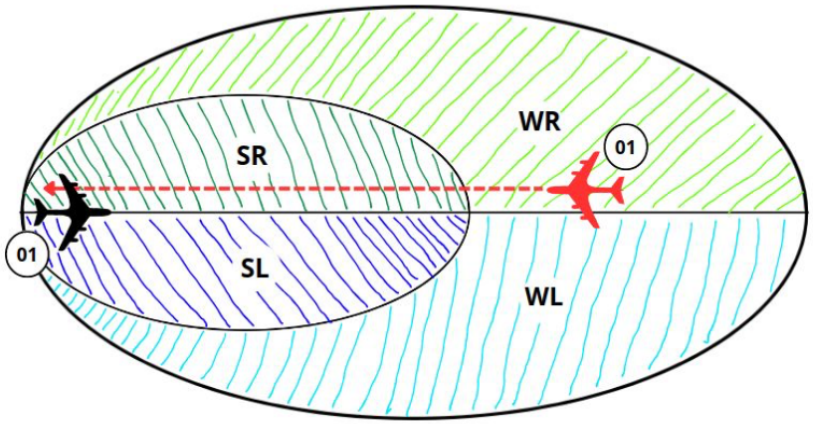
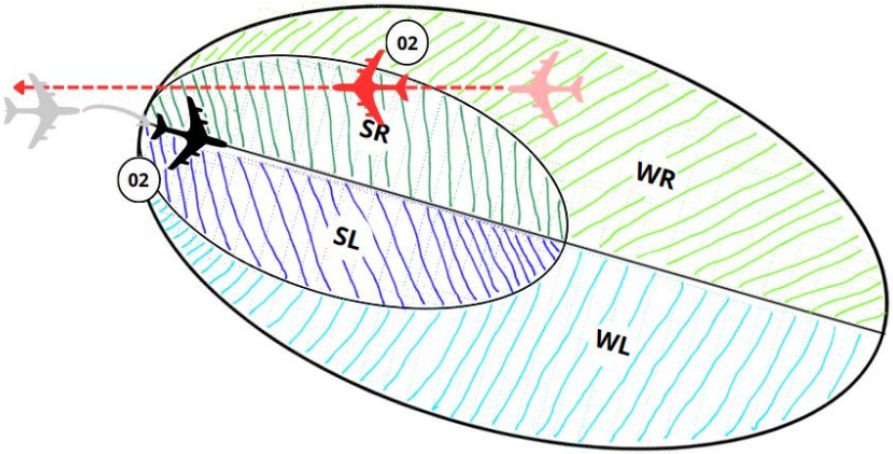
The "Lookahead" mechanism



	01		
Instantaneous Action	**WR**		

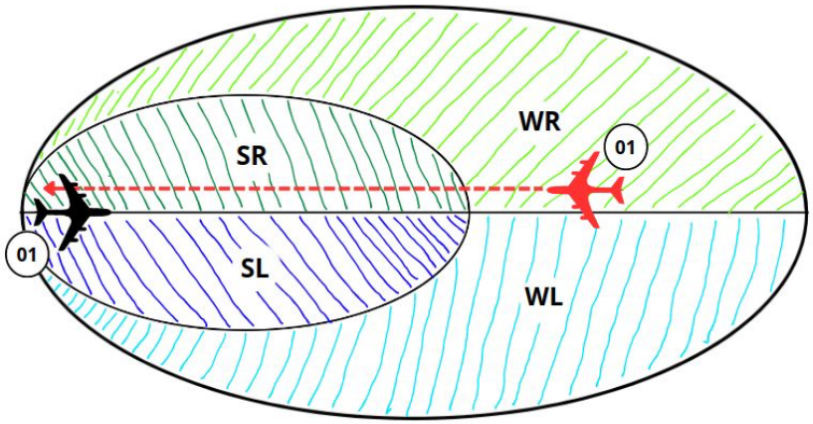
The "Lookahead" mechanism

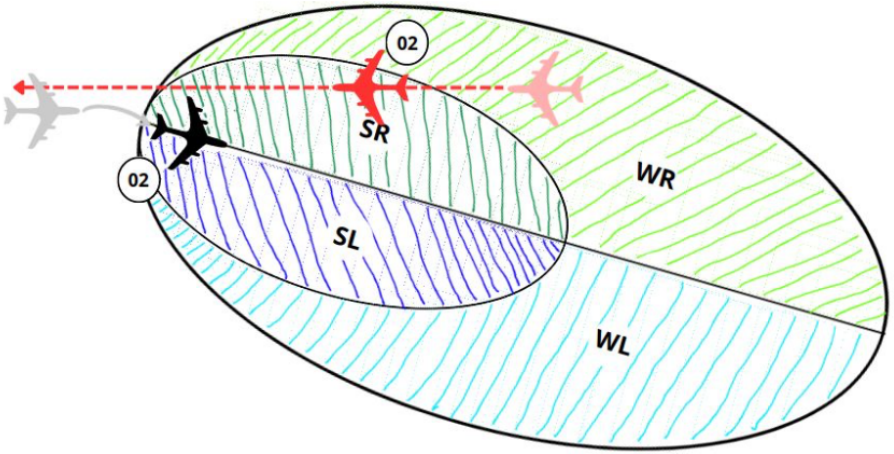


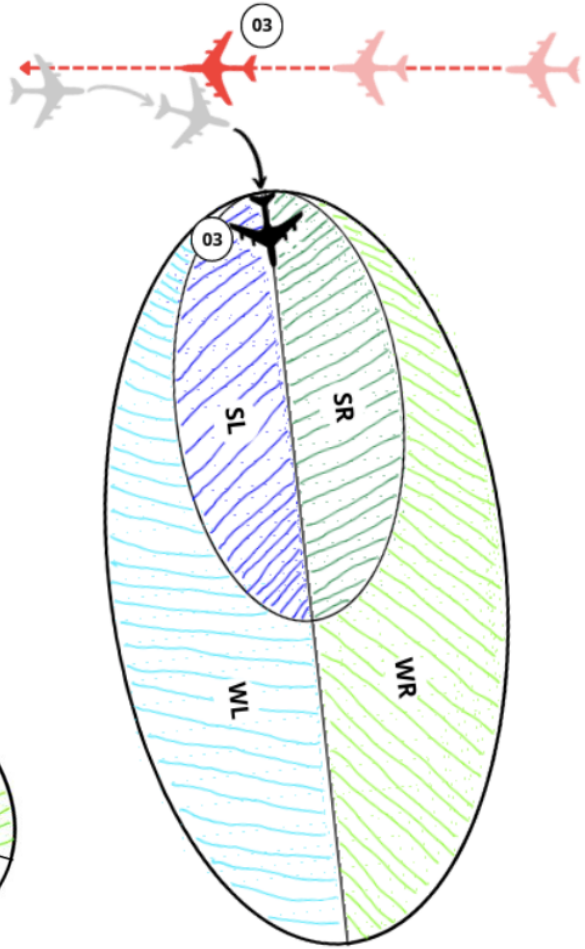


	01	02	
Instantaneous Action	**WR**	**SR**	

The "Lookahead" mechanism

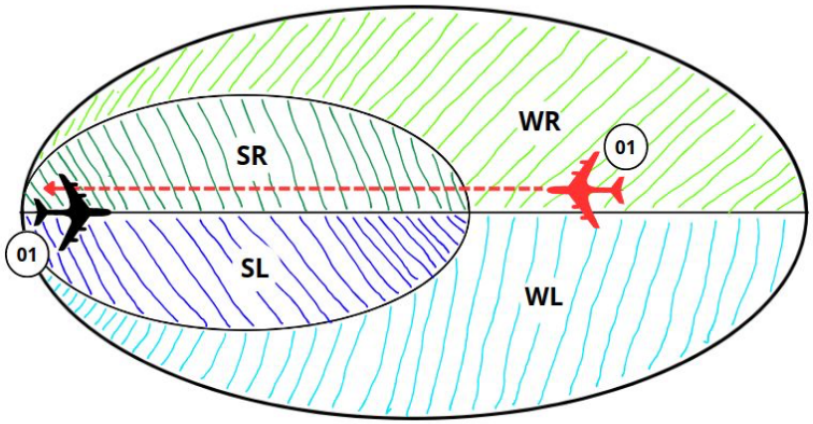


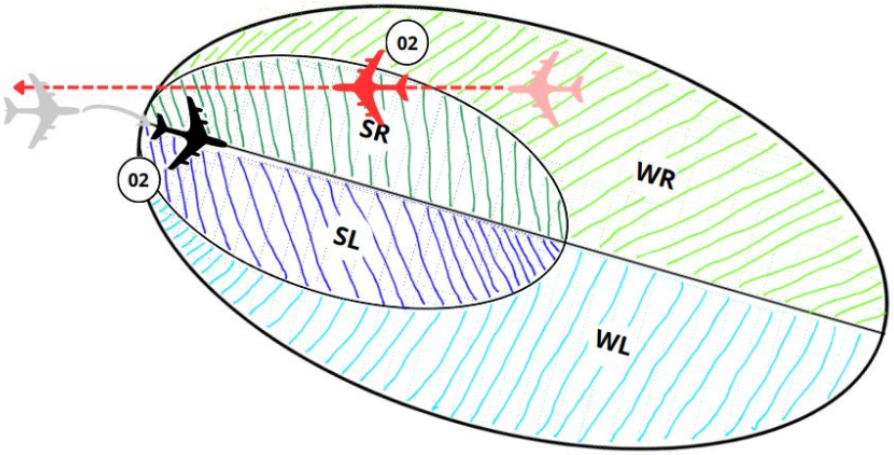


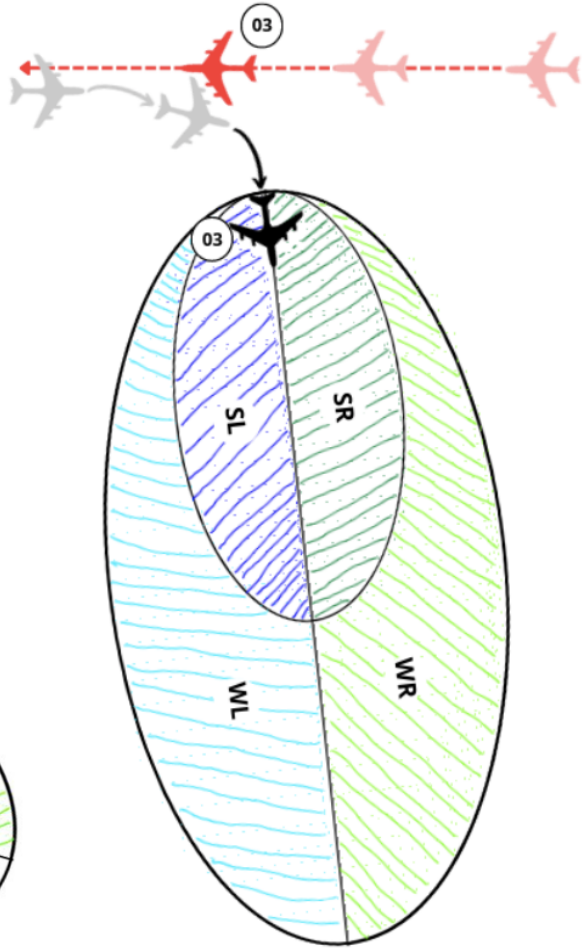


	01	02	03
Instantaneous Action	**WR**	**SR**	**COC**

The "Lookahead" mechanism







	01	02	03
Instantaneous Action	**WR**	**SR**	**COC**

Problems with the "Lookahead" mechanism

This "lookahead" mechanism is very complex but not enough explained in the literature...

It raises multiple questions :

- Why is there a lookahead for horizontal actions only ?
- How it is executed ?
- How long does it take to run it ?

We do not have any clear answers yet.

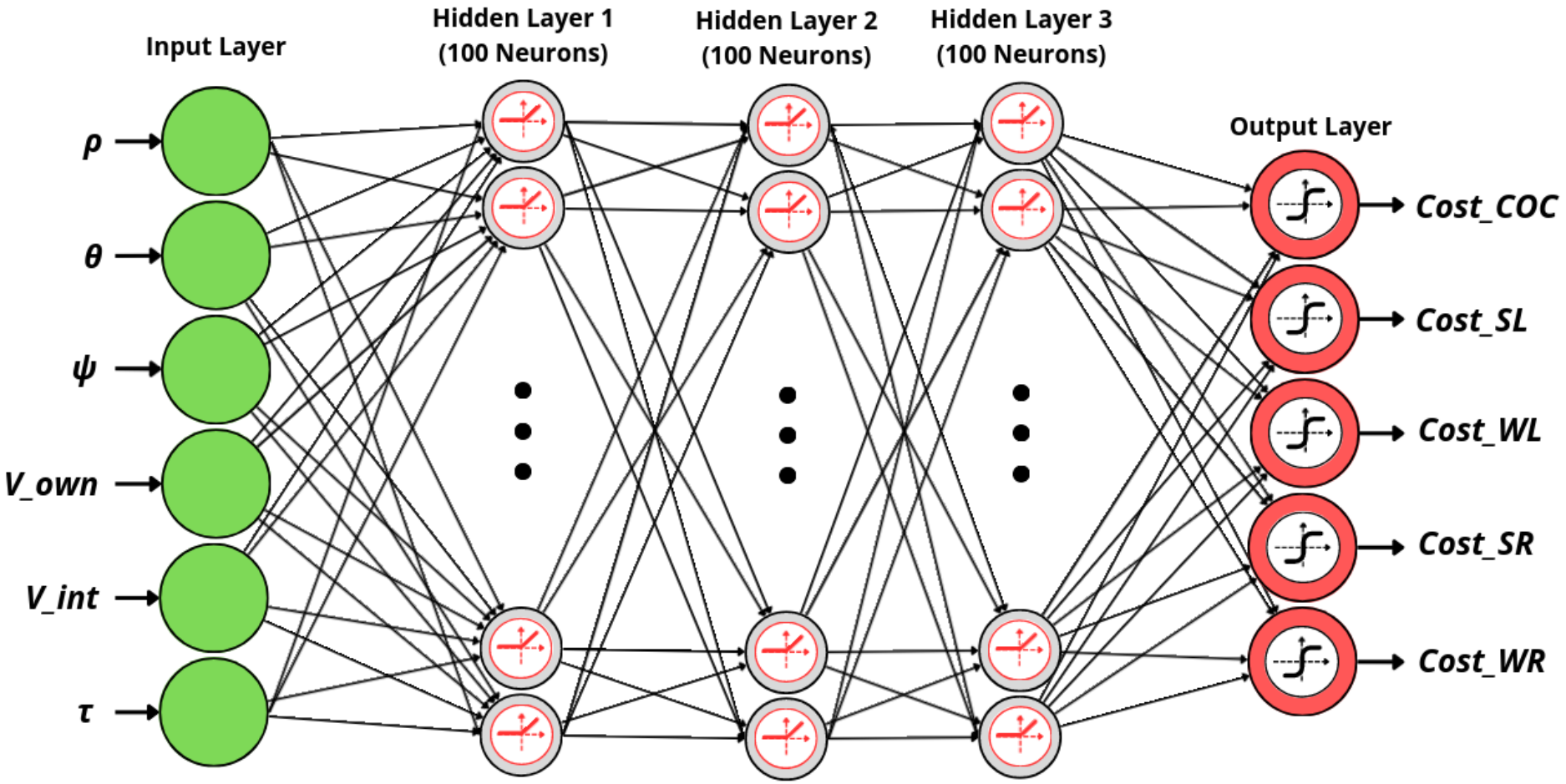
II - Approximations & Hypothesis for our work.

Approximations and Hypotheses

- We consider Horizontal plan problems only, but more complicated.
- We consider the simple case of "single intruder" for now.
- The ownship and the intruder are in a non-cooperative flight. There is no communication about their action.
- We don't consider the "lookahead" mechanism for now.

III - Our Neural Network.

Our Neural Network



Reference : Julian, K. D., Kochenderfer, M. J., & Owen, M. P. (2018). Deep Neural Network Compression for Aircraft Collision Avoidance Systems.

Work done so far

Learning process of the LUT for the Horizontal decisions.
Some intersting results so far :

- With 100 000 lines of the LUT :

```
Score Accuracy : 97.77699279785156
Ord Accuracy: 91.895
```

- With 500 000 lines of the LUT :

```
Score Accuracy : 86.52339935302734
Ord Accuracy: 81.283
```

- With 1 000 000 lines of the LUT :

```
Score Accuracy : nan
Ord Accuracy: 26.2955
```

Good results with few data used during the learning process.
But it gets worse by increasing the learning data.

Conclusion and future work

The learning process is still to be upgrade. Different architectures can still be tried to find the best neural network.

The learning process could be done on severals segments of the LUT divided depending on the "range".
This division could allow us to apply the "lookahead" mechanism ?

Once our functional neural network done, the objective will be to start formal verification of ACAS-Xu system using neural networks architectures.

Thank You ! !